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H19 overexpression drives an inflammatory response in mice

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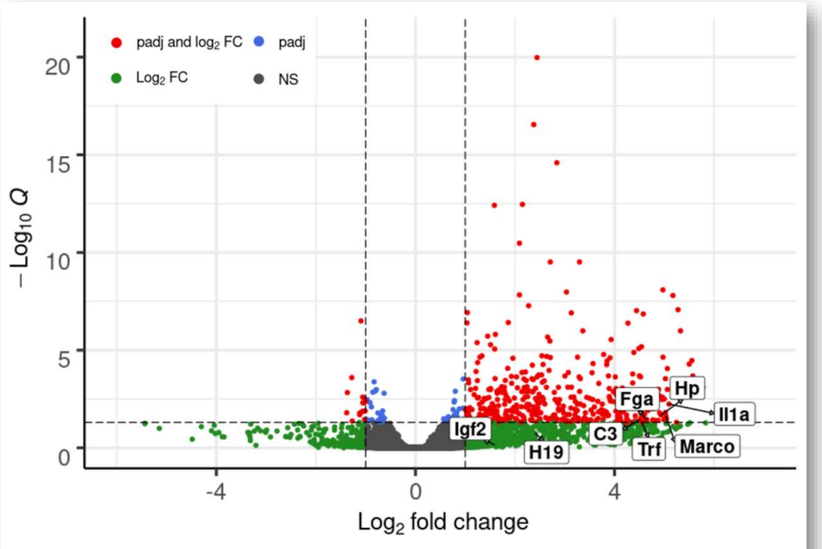
Introduction

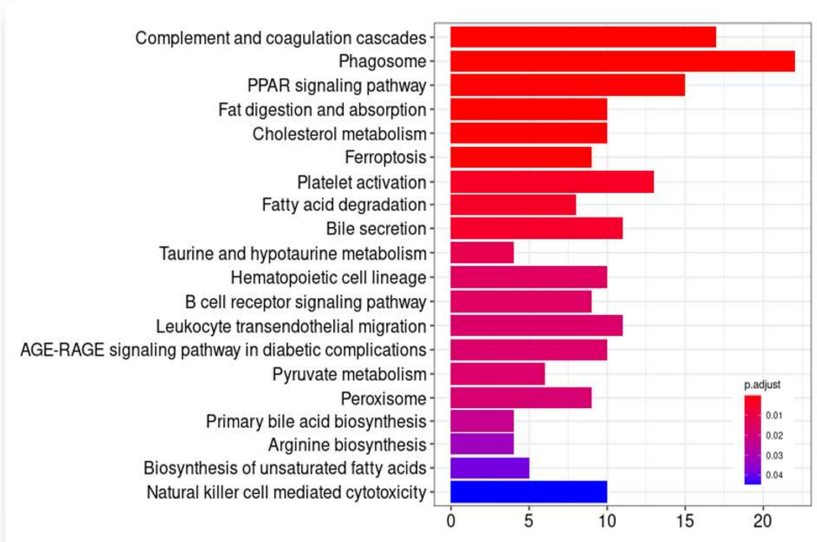
The *H19/IGF2* IG-DMR (IC1) regulates the reciprocal imprinting of *H19* and *Igf2*, and its abnormal methylation and genetic variants are frequently associated with the Silver-Russell syndrome (SRS) and Beckwith-Wiedemann syndrome (BWS). We have previously generated several transgenic mouse lines with humanized IC1 alleles, by replacing the endogenous mouse IC1 (mIC1) with orthologous human wildtype and mutant IC1 sequences [*1*,*2*]. In particular, the mice carrying a 2.2 kb deletion in the human IC1 (hIC1Δ2.2), found in BWS familial cases, mimic the growth and DNA methylation phenotypes and *H19/Igf2* expression changes that are typical of BWS if the mutation is maternally transmitted and those of SRS if the mutation is paternally transmitted. In order to investigate the physio-pathological mechanisms of the two syndromes, we have deeply characterized the molecular pathways altered in our mouse lines.



Volcano plot representing deregulated genes in neonatal kidney of hIC1Δ2.2pat mice

RNA-sequencing analysis

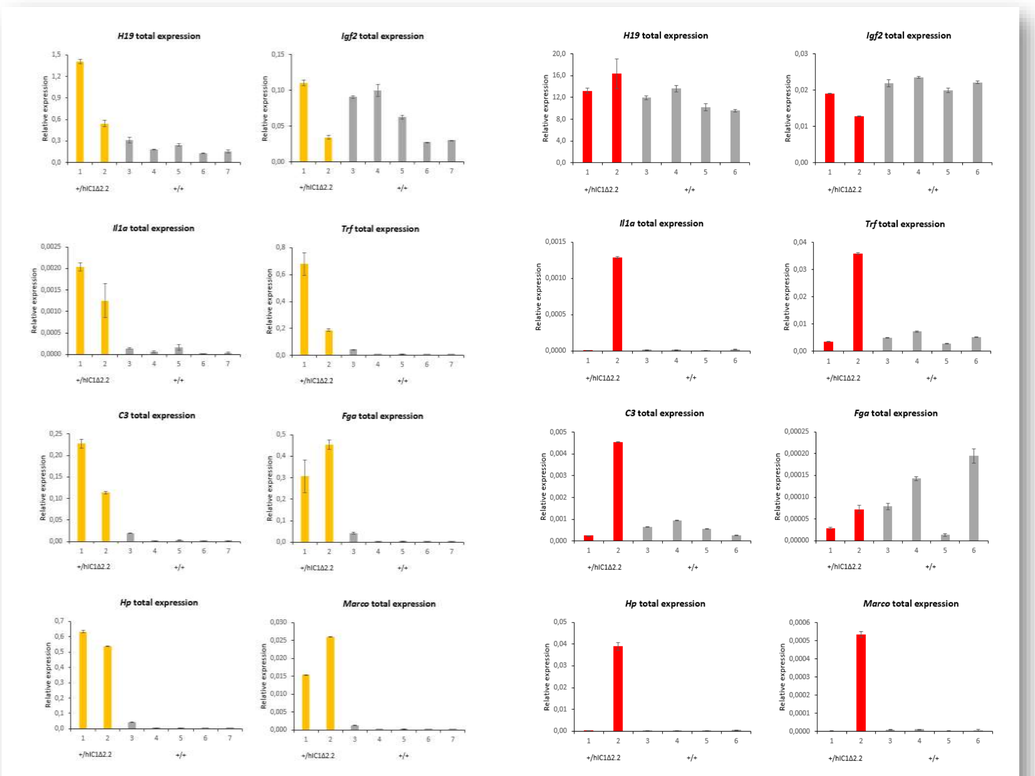
In the kidney of newborn mice (day 1) with paternal transmission of the transgene (hIC1Δ2.2pat), we found over 500 deregulated genes with respect to their controls. Interestingly, a large proportion of these genes including the most upregulated ones (e.g. Interleukin-1a, interleukin receptors and antagonists, macrophage receptors, complement factors, and coagulation factors) are involved in the inflammatory response.



Bar plot representing the 20 most significant pathways in neonatal kidney of hIC1Δ2.2pat mice

Evaluation of « inflammation genes » expression

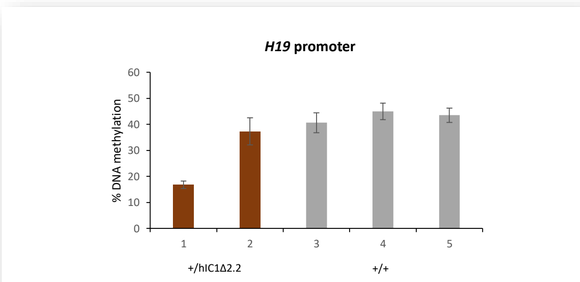
Il-1a, *Trf*, *C3*, *Marco*, *Fga*, *Hp* (here collectively referred to as "inflammation genes") have different functions (as cytokine, transport of iron in bloodstream, complement system, macrophage receptor, coagulation factor and haptoglobin precursor, respectively), but, if deregulated, they can be involved upstream or downstream of the inflammatory process. RT-qPCR analysis confirmed that most of them were strongly upregulated in the hIC1Δ2.2pat mice. However, we found that this effect was present only in 15% (3/20) of the transgenic animals analysed so far. Intriguingly, in these animals, the inflammatory response was found extended to other tissues (e.g. 1/6 in heart and 1/6 in liver)



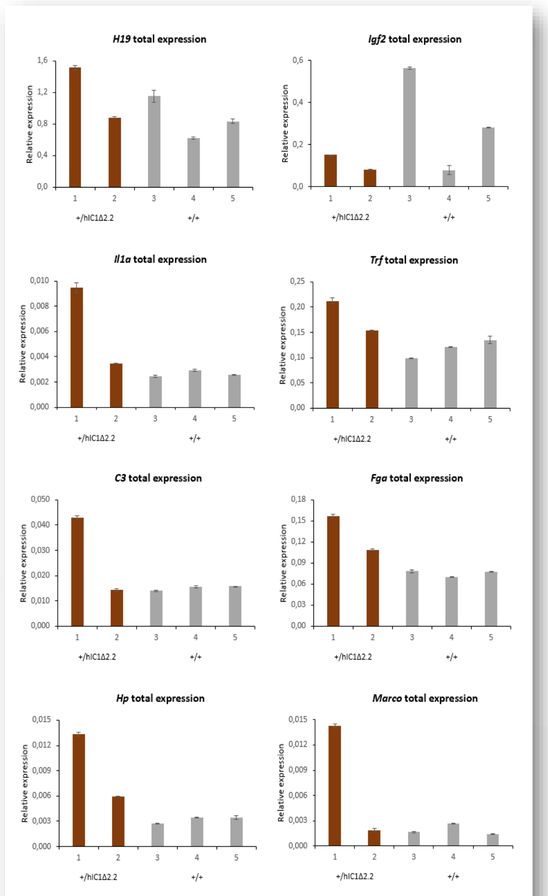
Gene expression analysis of representative kidney and heart samples of hIC1Δ2.2pat day 1 mice. Total expression of *H19*, *Igf2*, *Il-1a*, *Trf*, *C3*, *Marco*, *Fga* and *Hp were* assayed by RT-qPCR in neonatal kidney (*orange*) and neonatal heart (*red*) of individual hIC1Δ2.2pat mice, and compared to wild type littermates (*grey*). Bars represent each individual means ± SEM.

Correlation between inflammatory response and methylation of the *H19* promoter

We previously demonstrated that the hIC1Δ2.2pat mice show tissue-specific and mosaic imprinting alterations of *Igf2* and *H19* with partial loss of methylation (LOM) of the hIC1 region compared to their wild type controls [*2*]. Here, we found that the hIC1Δ2.2pat mice with activation of inflammatory response genes showed a more marked LOM of the *H19* promoter region, which correlated with a 1.5-2-fold higher expression of *H19*.



DNA methylation analysis of representative liver samples of hIC1Δ2.2pat mice. Percent methylation measured by pyrosequencing at *H19* promoter in embryonic (E15) liver of hIC1Δ2.2pat mice (brown), compared to wild type littermates (grey). Each histogram represents the methylation mean value of 6 CpGs. Bars represent each individual means ± SEM.



Gene expression analysis of representative liver samples of hIC1Δ2.2pat mice. Total expression levels of *H19*, *Igf2*, *Il-1a*, *Trf*, *C3*, *Marco*, *Fga* and *Hp* in embryonic (E15) liver (*brown*) of individual hIC1Δ2.2pat mice, compared to wild type littermates (*grey*), assayed by RT-qPCR. Bars represent each individual means ± SEM.

Conclusions

Strong activation of inflammatory response genes with respect to controls were found in several tissues of a small fraction (15%) of embryonal and neonatal hIC1Δ2.2pat mice. A correlation was found between the expression levels of the inflammatory response genes and that of *H19* but not with that of *Igf2*. *H19* activation is accompanied by a stronger hypomethylation of the *H19* promoter. We hypothesize that the hIC1Δ2.2 mutation leads to metastable methylation of the *H19* promoter and variegation of *H19* expression, and the mice with higher *H19* expression display the inflammatory response. Consistent with this hypothesis, several studies have involved the lncRNA-H19 in inflammatory processes [*3*-*8*]. Further analyses are in progress to determine the molecular pathways linking H19 with the inflammatory response genes in our mice.

References

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